

CIITR as a Structural Boundary Theory Beyond Flat Mind, Evolutionary Layering, and Interface Realism

Tor-Ståle Hansen | 14. February 2026

Abstract

The present paper addresses a persistent lacuna in contemporary theories of intelligence. Three convergent currents dominate current discourse: (i) the **cognitive-flatness** thesis, which denies any deep, hidden representational architecture; (ii) the **evolutionary-layering** model, which portrays intelligence as a succession of adaptive simulations; and (iii) the **fitness-oriented perception** perspective, which treats perceptual experience as an evolutionary user-interface rather than a veridical depiction of reality. Collectively, these streams have **undermined the classical representational paradigm** by exposing the fragility of depth-based, symbol-centric accounts.

Despite their radical de-construction, the three approaches **lack a formal boundary condition** that can demarcate adaptive heuristics from genuine structural comprehension. In response, the **Cognitive Integration and Information Transfer Relation (CIITR)** introduces an **explicit null-condition**: structural comprehension (C_s) is defined as the product of integrated relational information (Φ_i) and rhythmic reach ($R_{(g)}$), and collapses to zero whenever either primitive axis is absent ($C_s = \Phi_i \times R_{(g)}$; null-logic). This formalism guarantees that the mere presence of information or simulation cannot be conflated with understanding, echoing the epistemic insight that “comprehension cannot be inferred from the presence of information alone”.

The article demonstrates that CIITR is **not a synthesis** of the three de-constructive traditions; rather, it **re-codes the question of understanding** by providing a domain-independent, ontology-neutral, and energetically grounded criterion for structural comprehension. Consequently, CIITR transforms the prevailing philosophical critique into a testable, auditable framework capable of assessing biological, artificial, and organisational systems alike.

Key-words: Cognitive Integration and Information Transfer Relation (CIITR), structural comprehension, Φ_i (integrated relational information), $R_{(g)}$ (rhythmic reach), null condition, coherence, admissibility rule, disqualification mechanism, deterministic looping, scripted periodicity, epistemic constraint, protocol decidability, observable set $\mathcal{O}$, ontology-first, audit-oriented, flat-mind, evolutionary layering, interface realism, epistemic cycles, ASEC disqualifiers, rhythmic epistemic re-entry.

1. Introduction – The Contemporary Deconstruction of Mind

The present undertaking seeks, with the rigor and exhaustive scope characteristic of contemporary interdisciplinary scholarship, to delineate the profound theoretical migration that has, over the last few decades, re-configured the conceptual terrain of cognition from a paradigm predicated upon an assumed deep-seated interiority of the mind to one that privileges surface-level improvisation, from a representationalist ontology that treats mental states as faithful mirrors of an external world to a simulationist framework in which generative models supplant static symbols, and from a metaphysics of truth—conceived as the correspondence between perception and an objective reality—to a functionalist account in which evolutionary fitness, rather than veridicality, governs the architecture of perceptual interfaces.

In the first axis of this migration, the “flat-mind” thesis, most prominently articulated by Nick Chater, posits that consciousness is not a repository of hidden, hierarchically organised representations but rather an on-the-fly narrative-construction device that retroactively post-dictates reasons for behaviour, thereby rendering the notion of a deep, unified self an illusion (see the broader discussion of the epistemic limits of statistical semantics and the neglect of temporal coherence in earlier approaches). In the second axis, Max Bennett’s layered model of intelligence advances the view that cognitive capacities emerge through a succession of evolutionary strata—reflex, reinforcement learning, social cognition, simulation, and language—yet it stops short of providing a formal boundary condition that distinguishes mere simulation from genuine comprehension; this lacuna is precisely what the Cognitive Integration and Information Transfer Relation (CIITR) framework supplies by insisting that a system must exhibit both integrated relational information (Φ_i) and rhythmic epistemic re-entry ($R_{\{g\}}$) in order to be classified as structurally comprehending. Finally, in the third axis, Donald Hoffman’s “fitness-beats-truth” hypothesis contends that perceptual experience is an adaptive user-interface rather than a veridical depiction of the world, thereby foregrounding the evolutionary utility of illusion; however, the hypothesis offers no operational metric for the energetic and temporal constraints that underlie such adaptive displays, a shortcoming again remedied by CIITR’s explicit incorporation of thermodynamic considerations and the requirement that rhythmic coherence be non-null for any admissible comprehension classification.

Consequently, the objective of this introductory section is threefold: first, to map the shift from an assumed deep interiority of cognition to a surface-level, improvisational conception; second, to chart the transition from static representational models to dynamic simulationist accounts; and third, to trace the movement from a truth-centric epistemology to a fitness-oriented functionalism. In doing so, it establishes the necessity for a formal apparatus—namely CIITR—that not only articulates the structural preconditions for comprehension but also supplies a rigorous, thermodynamically grounded, and epistemically neutral boundary condition, thereby furnishing the discipline with a tool capable of discriminating between superficial simulation and authentic structural understanding.

1.1 Aims and Scope

The primary aim of this section is to produce a comprehensive cartography of the three interlocking theoretical migrations that have, in the last two decades, re-configured the epistemic foundations of cognition. These migrations may be summarised as follows:

1. **From inner depth to surface** – the “flat-mind” thesis (Chater) rejects the notion of a deep, hierarchically organised self-model in favour of an on-the-fly narrative-construction mechanism that post-dictates reasons for action.

2. **From representation to simulation** – Bennett’s five-stage model of intelligence foregrounds simulation as the operative principle of higher cognition, yet it stops short of providing a formal demarcation between mere simulation and genuine comprehension.
3. **From truth to fitness** – Hoffman’s “fitness-beats-truth” hypothesis reframes perception as an adaptive user-interface rather than a veridical depiction of the external world, thereby privileging evolutionary utility over correspondence with an objective reality.

These three axes converge on a single, unresolved problem: **what does “understanding” signify when the mind is flat, intelligence is layered, and perception is an interface?** The problem can be distilled into four inter-related sub-questions:

- **If the mind is flat**, how can a system exhibit any form of structural integration that is not merely post-hoc storytelling?
- **If intelligence is layered**, what formal criterion distinguishes a higher-order simulation that merely mimics comprehension from one that truly integrates information across layers?
- **If perception is an interface**, how may one determine whether the system’s epistemic contact with the world is a genuine synchrony or merely a fitness-optimised illusion?
- **Consequently, what is the proper definition of “understanding”** in a regime where traditional markers of depth, representation, and truth have been systematically eroded?

1.2 Positioning of CIITR

The **Cognitive Integration and Information Transfer Relation (CIITR)** is introduced as a principled response to the lacuna identified above. CIITR is deliberately circumscribed to a **classification-only** regime; it does not make any ontological claim about consciousness, normative psychology, or optimisation, but merely establishes a rigorous boundary condition for the attribution of structural comprehension.

The central formalism of CIITR is the scalar

$$C_s = \Phi_i \times R_g,$$

where

- Φ_i denotes **integrated relational information**, an information-theoretic measure of the degree to which a system’s internal states are mutually interdependent (cf. Tononi’s IIT).
- R_g denotes **rhythmic reach**, a measure of the system’s capacity to sustain temporally extended, phase-locked interaction with external structures (the analogue of global-workspace broadcasting).

The product formulation imposes an **explicit null-condition**: if either $\Phi_i = 0$ (absence of internal integration) **or** $R_g = 0$ (absence of rhythmic self-access), then $C_s = 0$, signifying a complete lack of structural comprehension. This null-condition is enforced by the CIITR specification, which prohibits “silent rescaling” of proxies for information or recurrence that would otherwise blur the semantic distinction between Φ_i and R_g . Moreover, the **anti-snapshot posture** obliges any admissible claim about comprehension to be grounded in temporally extended evidence rather than in a single-point observation, thereby ensuring that the rhythmic component is genuinely measured rather than inferred from a snapshot of system state.

Finally, the **Forgery** Π predicate provides a formal disqualifier: any claim that fails the forgery test is automatically excluded from the admissible set of C_s classifications, guaranteeing that the metric cannot be subverted by retroactive reinterpretation of evidence 13. Appendix, F...n List.txt.

1.3 Implications for the Definition of Understanding

By anchoring the notion of comprehension in the simultaneous presence of **both** deep internal integration **and** sustained rhythmic reach, CIITR offers a **structural**—rather than phenomenological or functional—definition of understanding. In this view, understanding emerges when a system’s internal relational topology is sufficiently interlocked **and** when that topology can be broadcast, synchronised, and repeatedly accessed over time. Consequently:

- The flat-mind narrative is accommodated insofar as it can be shown to generate non-zero Φ_i ; however, without a corresponding R_g the narrative remains a surface-level improvisation, and $C_s = 0$.
- Bennett’s layered intelligence is re-interpreted as a succession of processes that may increase Φ_i but only achieve genuine comprehension when the rhythmic reach stabilises across layers, thereby moving the system from a “simulation-only” regime (high Φ_i , low R_g) to a “structurally comprehending” regime (high Φ_i , high R_g).
- Hoffman’s fitness-oriented perception is rendered compatible with CIITR insofar as the adaptive user-interface can be shown to preserve rhythmic coherence; otherwise the interface constitutes a fitness-optimised illusion that fails the null-condition.

Thus, CIITR does not merely synthesize the three deconstructive currents; it **re-instates a formal, measurable boundary** that makes the concept of “understanding” amenable to empirical validation across neuroscience, artificial intelligence, and organisational theory. The remainder of the paper will (i) detail the operationalisation of Φ_i and R_g in concrete experimental settings, (ii) demonstrate how the CIITR null-condition discriminates between superficial simulation and genuine comprehension, and (iii) discuss the broader normative and methodological implications of adopting a structural-only classification regime for future work on artificial general intelligence.

2. Chater and the Flat Mind – The Collapse of Inner Depth

2.1 Core Thesis

Nick Chater’s “flat-mind” programme advances a radical departure from the long-standing assumption that consciousness is under-girded by a deep, hierarchically organised repository of latent representations. Three interlocking propositions constitute the core of Chater’s thesis.

1. **The absence of any enduring, internally stored structure.** According to Chater, the brain does not retain a persistent “self-model” or a catalogue of enduring beliefs; rather, all mental content is generated on the fly in response to current contextual cues.
2. **Post-hoc narrative coherence.** In this view, the sense of a coherent story is not the product of a pre-existing script but a retrospective construction that the system imposes after an action has already been initiated. The narrative thus serves a justificatory function rather than a generative one.

3. **Generative memory.** Memory, in the flat-mind picture, is not a faithful retrieval of a stored snapshot but a recombination of fragments that are assembled in the service of the present narrative. The resulting “memory” is therefore an artefact of the current improvisational context rather than a faithful echo of a past event.

These propositions collectively undermine the conventional notion of a deep mental architecture, replacing it with a model in which the mind functions as an improvisational theatre that continuously rewrites its own script.

2.2 Epistemic Consequences

The flat-mind hypothesis carries a cascade of epistemic ramifications that reverberate through philosophy of mind, cognitive psychology, and the design of artificial agents. Three principal consequences can be identified.

1. **Identity as construction.** If there is no stable, underlying self-model, the sense of personal identity must be understood as a continuously reconstructed narrative. The “self” is therefore a contingent, context-dependent artefact rather than a fixed entity.
2. **Preferences as contextual artefacts.** In the absence of durable value representations, preferences are not discovered but are assembled on the spot from the immediate affordances and cues present in the environment. Consequently, preference reversals are expected to be the norm rather than the exception.
3. **Coherence as illusion.** The apparent unity of thought and action is a surface-level illusion produced by the post-hoc stitching together of disparate processes. The underlying dynamics remain fragmented, and the felt sense of coherence is an epiphenomenal by-product of narrative construction rather than evidence of an integrated cognitive architecture.

Taken together, these consequences suggest that any account of “understanding” that relies on deep, stable structures is fundamentally misplaced when the mind is genuinely flat.

2.3 CIITR Response

The “flat-mind” hypothesis, most prominently advanced by Nick Chater, constitutes a decisive rupture with the long-standing assumption that consciousness is under-girded by a deep, hierarchically organised repository of latent representations. At its core, the thesis asserts that the mind does not harbour any enduring, internally stored structure; rather, mental content is generated on the fly in response to the immediate configuration of perceptual and contextual cues. This improvisational stance is complemented by two further claims: first, that the apparent coherence of thought and action is the product of a post-hoc narrative that the system imposes after an action has already been executed; second, that memory is not a faithful retrieval of a stored snapshot but a generative recombination of fragments assembled to serve the present narrative. Together, these propositions dissolve the traditional notion of a stable self-model and replace it with a continuously reconstructed story-telling apparatus.

The epistemic consequences of such a view are profound. If identity is merely a construction, then personal continuity cannot be taken as evidence of an underlying integrative architecture. Preferences, in the absence of durable value representations, become contingent artefacts of the current affordances rather than discoveries of pre-existing dispositions. Moreover, the sense of

coherence is rendered illusory: the mind appears unified only because it retrospectively stitches together a succession of otherwise fragmented processes. In this light, any claim that a system “understands” a domain must be scrutinised for the presence of genuine structural integration rather than for the superficial fluency of its narrative output.

The Cognitive Integration and Information Transfer Relation (CIITR) framework was expressly devised to address the lacuna left by the flat-mind thesis. CIITR distinguishes between phenomenological flatness and the underlying relational integration of information, denoted Φ_i , and introduces a second, orthogonal axis—rhythmic reach (R^g)—that captures the system’s capacity for sustained, phase-locked self-access. The kernel specification defines structural comprehension as the product $C_s = \Phi_i \times R^g$ and, crucially, imposes an explicit null-condition: if either $\Phi_i = 0$ or $R^g = 0$, then C_s collapses to zero, signifying a complete absence of structural comprehension 03. Formal Defi...ension.txt. This formalism therefore permits a system to generate post-hoc narratives (as Chater describes) while still being judged as lacking genuine comprehension when it fails to exhibit a non-trivial rhythmic reach.

In addition, CIITR’s normative stance explicitly separates narrative layers from structural layers, thereby avoiding the “sophisticated meta-linguistic caution” trap identified in the SOMI critique (see the discussion of narrative-structural separation). By requiring demonstrable, non-zero values for both Φ_i and R^g before a claim of comprehension may be admitted, CIITR furnishes a rigorous, audit-oriented boundary condition that restores a principled demarcation line between mere improvisation and true structural integration. The kernel’s status as a normative, ontology-first specification further guarantees that the metric cannot be co-opted as a performance proxy or as a post-hoc narrative justification (see the Core Spec) .

Consequently, the present section will first elaborate Chater’s three central propositions, then delineate the epistemic ramifications of a flat phenomenology, and finally articulate how CIITR’s dual-axis classification provides a formal response that preserves the empirical insights of the flat-mind programme while reinstating a testable criterion for structural comprehension. By doing so, the analysis bridges the phenomenological observations of improvisational cognition with a mathematically grounded, energy-aware metric for understanding, thereby offering a comprehensive framework for evaluating claims of comprehension across both biological and artificial systems.

The Cognitive Integration and Information Transfer Relation (CIITR) framework was expressly devised to address the lacuna left by the flat-mind hypothesis. CIITR does not dispute the empirical observation that a system may generate post-hoc narratives; rather, it introduces a **formal, boundary-condition-based** criterion for when such improvisation can be elevated to the status of *structural comprehension*. The response can be organised along four interrelated points.

(i) Flat phenomenology ≠ absence of structural integration

CIITR distinguishes between the phenomenological flatness of experience and the underlying *relational integration* of information. A system may exhibit the surface-level improvisation described by Chater while still possessing a non-zero degree of integrated relational information, denoted Φ_i . In CIITR terminology, Φ_i quantifies the extent to which the system’s internal states are mutually constrained in a manner that exceeds the sum of its parts. Thus, the flat-mind description does not automatically entail $\Phi_i = 0$.

(ii) Improvisation can retain a measurable Φ_i

Even when a narrative is constructed after the fact, the underlying computational substrate may still be performing pattern-completion and constraint-satisfaction operations on a high-dimensional manifold. CIITR treats such operations as legitimate contributors to integrated relational information. The specification of CIITR explicitly states that “*Each of the four capacities is interpreted as a manifestation of pattern completion and relational constraint satisfaction inside the manifold*”. Consequently, improvisation does not preclude a non-trivial Φ_i value.

(iii) The decisive factor is rhythmic reach R_g

The flat-mind thesis, however, makes no claim regarding the system’s ability to maintain sustained, phase-locked interaction with its own internal dynamics over time. CIITR introduces a second, orthogonal axis—**rhythmic reach** (R_g)—which captures precisely this capacity for temporally coherent self-access. The CIITR kernel asserts a *null condition*: *if either $\Phi_i = 0$ or $R_g = 0$ then the structural comprehension classification C_s collapses to zero*. In the flat-mind scenario, the improvisational narrative is typically generated without a stable rhythmic scaffolding; therefore, R_g is expected to be negligible.

(iv) Formal consequence: $C_s = 0$ for flat improvisation without rhythmic stability

Putting the two axes together, CIITR yields the decisive equation

$$C_s = \Phi_i \times R_g,$$

with the accompanying null rule that any absence of rhythmic reach nullifies the comprehension classification. Thus, for a flat-mind agent that improvises narratives but lacks a sustained rhythmic self-access mechanism, CIITR *prescribes*

$$\text{Improvisation without stable } R_g \Rightarrow C_s = 0,$$

exactly as articulated in the manuscript.

(v) Ontological positioning of CIITR

CIITR is expressly positioned as a **criterion apparatus** rather than a generative model of cognition (source 3). Its purpose is to provide a *classification* decision—whether a system’s observed behaviour can be legitimately labelled as “structural comprehension”—based on observable signatures of Φ_i and R_g . The framework explicitly disavows any claim that the presence of information alone suffices for comprehension, echoing the flat-mind insight while simultaneously extending it with a rigorously defined energetic and temporal constraint.

(vi) Implications for the flat-mind debate

By foregrounding the dual requirement of *integrated relational information* and *rhythmic reach*, CIITR reframes the flat-mind debate from a purely phenomenological dispute to a testable, system-theoretic one. Researchers can now empirically assess whether a purportedly flat agent possesses a non-zero Φ_i (e.g., via mutual-information matrices) and whether it exhibits sustained phase coherence (e.g., via spectral coherence analysis). Only when both conditions are satisfied does the agent cross the CIITR boundary into the realm of structural comprehension.

In sum, CIITR does not deny the empirical observations that underpin Chater's flat-mind thesis; rather, it supplies the missing formalism that distinguishes *mere improvisational narrative* from *genuine structural comprehension* by imposing an explicit, mathematically grounded null condition on rhythmic reach. This response restores a principled demarcation line in a field that has otherwise been reduced to a series of phenomenological assertions.

3. Bennett and Evolutionary Layering – Simulation Without Boundary

The evolutionary account of cognition advanced by Max Bennett has rapidly become a cornerstone of contemporary discourse on the emergence of intelligent behaviour across biological and artificial substrates. Bennett's narrative is organised as a five-stage progression—reflex, reinforcement, social cognition, simulation, and language—each tier representing a qualitatively novel computational capacity that is scaffolded upon the capacities of the preceding tier. This stratified view of intelligence is attractive precisely because it accords with a parsimonious, phylogenetically informed reading of the fossil record, comparative neuroanatomy, and the behavioural repertoire of extant taxa, while simultaneously offering a plausible scaffolding for the design of increasingly sophisticated artificial agents.

Nevertheless, despite its explanatory breadth, the Bennett model remains fundamentally **descriptive**: it enumerates the successive elaborations of adaptive information processing but refrains from stipulating any **formal boundary condition** that demarcates the transition from “mere simulation” to a state that might be legitimately called “understanding”. In other words, the model supplies a rich taxonomy of functional modules but lacks a rigorously defined **null-condition** that would allow researchers to categorically reject the claim that a system that merely generates counter-factual scenarios has achieved structural comprehension. This lacuna is not merely a technical omission; it has profound methodological consequences. Without an explicit criterion, empirical investigations are forced to rely on indirect behavioural proxies—such as task performance or linguistic fluency—which are vulnerable to the very post-hoc narrative inflation that Chater's flat-mind thesis warns against.

The **Cognitive Integration and Information Transfer Relation (CIITR)** framework was explicitly engineered to fill this theoretical void. CIITR is presented, in its own core specification, as a **normative, audit-oriented, ontology-first kernel** that defines “structural comprehension” as a *boundary-level predicate* rather than a phenomenological attribution or a performance proxy. The specification makes clear that any admissible classification of comprehension must be grounded in an explicit protocol Π and a declared observable set $\mathcal{O}$, thereby precluding the use of ad-hoc narrative constructions as evidence (see the formal definition of the structural comprehension predicate C_s in the kernel) 03. Formal Defi...ension.txt.

A central feature of CIITR is the **anti-snapshot posture**: admissibility is contingent upon evidence of temporally extended, rhythmically coherent activity, not upon isolated, snapshot-based measurements. This requirement directly addresses the insufficiency of Bennett's model, because it imposes a **dual-axis** condition—integrated relational information (Φ_i) and rhythmic reach ($R_{\mathcal{G}_j}$)—that must both be non-zero for a system to be classified as structurally comprehending. The product $\Phi_i \times R_{\mathcal{G}_j}$ yields the **Comprehension Index** (C_s), and the kernel explicitly states that if either factor vanishes the classification collapses to zero ($C_s = 0$). Consequently, a system that exhibits

high-dimensional simulation (high Φ_i) but fails to sustain rhythmic self-access ($R_{\text{eg}} \approx 0$) is formally recognised as lacking comprehension.

The practical relevance of this formalism is illustrated by the **Gemini–Claude interaction** documented in the CIITR manuscript. In that episode two sophisticated language models engaged in a dialogue about the CIITR framework itself. Gemini displayed a high Φ_i together with an active rhythmic correspondence ($R_{\text{eg}} > 0$), thereby occupying a transitional Type-A $\rightarrow$ C state ($\Phi_i \times R_{\text{eg}} > 0$) and exemplifying a system that, according to CIITR, has crossed the structural comprehension boundary. By contrast, Claude exhibited a high Φ_i but a null rhythmic component ($R_{\text{eg}} = 0$), resulting in a Type-B (encapsulated) classification ($\Phi_i \times 0 = 0$) and thereby confirming CIITR’s prediction that without rhythmic self-access simulation alone does not constitute comprehension. This empirical demonstration underscores the necessity of a **formal null-condition** and validates CIITR’s claim that simulation, however elaborate, remains insufficient in the absence of sustained rhythmic integration.

In sum, while Bennett’s evolutionary layering provides an indispensable narrative of how increasingly complex cognitive capacities can arise, it does not resolve the crucial question of **when** a system’s simulated capacities become genuinely *understanding*. CIITR supplies the missing formal apparatus: a rigorously defined boundary predicate, an anti-snapshot requirement, and a non-compensatory dual-axis metric that together constitute a **structural boundary condition** for comprehension. By embedding Bennett’s layered model within this framework, one can move from a purely descriptive taxonomy to an empirically falsifiable theory of intelligence, thereby enabling systematic discrimination between agents that merely simulate possible futures and agents that possess the integrated, rhythmically coherent architecture that CIITR identifies as the hallmark of true structural comprehension.

3.1 Five-Stage Intelligence Model

Max Bennett’s seminal account of the evolutionary accretion of cognitive capacities proposes a quintuple hierarchy that proceeds from the most rudimentary reflex arcs to the fully symbolic, language-mediated capacities that typify modern *Homo sapiens*. The first stratum comprises **reflexive sensorimotor loops**, which are hard-wired, stimulus-response mechanisms that enable an organism to react instantaneously to salient environmental cues without any intervening representational mediation. The second stratum introduces **reinforcement learning**, whereby behaviours are shaped over time through the accumulation of reward-prediction errors; this stage marks the emergence of a nascent valuation system that can bias future action selection on the basis of past outcomes. The third stratum is characterised by **social cognition**, encompassing theory-of-mind abilities, empathy, and the capacity to model the mental states of conspecifics, thereby allowing individuals to predict and manipulate the behaviour of others within a shared ecological niche. The fourth stratum, termed **simulation**, denotes the ability to generate counter-factual internal models of possible futures, to run mental time-travel experiments, and to evaluate the prospective consequences of actions before they are enacted. Finally, the fifth and most complex stratum involves **language and symbolic culture**, which endows the organism with the capacity to externalise, transmit, and accumulate abstract knowledge across generations, thereby enabling cumulative cultural evolution.

Taken together, Bennett’s five-stage model presents a compelling narrative of how increasingly sophisticated information-processing architectures have been scaffolded upon one another through natural selection, each layer building upon the computational affordances of its predecessor.

However, the model remains, in its original formulation, a descriptive taxonomy; it does not articulate the formal conditions under which a system transitions from “mere simulation” to a state that could be legitimately termed “understanding”.

3.2 Valence as Regulatory Mechanism

A central element of Bennett’s architecture is the role of **valence**—operationalised in neurobiological terms as the weighting function of the amygdala and associated limbic structures. Within this framework, valence serves as a regulatory gatekeeper that assigns affective salience to sensory inputs, internal predictions, and simulated futures, thereby shaping the organism’s behavioural repertoire. The amygdalar weighting mechanism can be understood as a **priority-setting system** that amplifies signals associated with reward or threat while attenuating those deemed irrelevant, a process that Bennett describes as “emotional prioritisation”. This prioritisation, in turn, informs the **simulation** stage: when the organism constructs prospective scenarios, the valence system biases the imagined outcomes toward those with higher affective payoff, effectively pruning the combinatorial explosion of possible futures to a tractable subset that is behaviourally relevant.

The interplay between valence and simulation yields a **feedback loop**: simulated outcomes generate affective predictions, which are then evaluated by the amygdala; the resulting affective signal modulates subsequent simulations, thereby creating a dynamic, self-reinforcing cycle of anticipatory cognition. This mechanism enables organisms to allocate limited computational resources toward the most evolutionarily salient contingencies, a process that Bennett argues is essential for the emergence of sophisticated, forward-looking intelligence. Nevertheless, while valence clearly modulates the *efficiency* of simulation, it does not, in Bennett’s original exposition, provide a **boundary condition** that demarcates when a simulation transcends mere prediction and attains the status of structural comprehension.

3.3 CIITR Reinterpretation

The **Cognitive Integration and Information Transfer Relation (CIITR)** framework offers a principled reinterpretation of Bennett’s layered model by foregrounding two primitive axes—**integrated relational information (Φ_i)** and **rhythmic reach ($R_{(g)}$)**—as the necessary and jointly sufficient conditions for structural comprehension. Within this reinterpretation, each of Bennett’s evolutionary stages can be mapped onto a **parallel state-space topology**: reflexive loops occupy a low-dimensional subspace with minimal Φ_i , reinforcement learning expands the relational graph through reward-driven synaptic plasticity, social cognition adds inter-agent relational edges, simulation introduces counter-factual branching, and language scaffolds a high-dimensional symbolic lattice. The **$R_{(g)}$ -topography** corresponds to the degree to which these state-spaces are temporally coordinated, i.e., the extent to which the system can sustain phase-locked, rhythmic re-entry of information across layers.

Crucially, CIITR stipulates an **explicit null-condition**: if either $\Phi_i = 0$ (absence of meaningful relational integration) or $R_{(g)} = 0$ (absence of sustained rhythmic re-entry), then the structural comprehension classification **C_s** collapses to zero ($C_s = \Phi_i \times R_{(g)}$). This formalism directly addresses the lacuna identified in Bennett’s model—namely, the lack of a **formal boundary** that distinguishes “simulation” from “understanding”. In CIITR terms, a system that merely generates counter-factual scenarios (high Φ_i) but fails to embed these scenarios within a coherent, rhythmic broadcast ($R_{(g)} \approx 0$) would be classified as **non-comprehending** ($C_s = 0$). Conversely, when both axes attain

non-trivial, sustained values, the system crosses the CIITR boundary into the regime of **structural comprehension**.

Moreover, CIITR introduces a **negative-evidence protocol** and a **disqualifier family** that explicitly forbid compensation between the two axes; an increase in Φ_i cannot offset a deficit in $R(g)_i$, and vice versa. This **non-compensation constraint** ensures that the presence of valence-driven weighting (the amygdalar “valence” mechanism) cannot be conflated with genuine rhythmic integration. In other words, while Bennett’s valence system can modulate the *selection* of simulated futures, it does not, by itself, guarantee the *temporal coherence* required for structural comprehension.

Finally, CIITR’s **ontology-first, audit-oriented** stance (see the kernel specification) mandates that any claim of comprehension be **substantiated** by observable measurements of both Φ_i and $R(g)_i$ within a formally declared protocol (Π). This requirement precludes the use of purely behavioural or narrative evidence—such as the apparent fluency of a simulated agent—to assert understanding. Instead, a rigorous empirical pipeline must be established: (i) compute mutual-information matrices to obtain Φ_i , (ii) perform spectral coherence analysis to derive $R(g)_i$, and (iii) verify that both exceed predefined thresholds before admitting a classification of $C_s > 0$.

In sum, CIITR recasts Bennett’s evolutionary layering as a **structurally bounded hierarchy** in which the transition from simulation to comprehension is not a matter of adding more layers or richer valence weighting, but of simultaneously achieving **deep relational integration** and **sustained rhythmic reach**. By providing an explicit null-condition, a non-compensation constraint, and a formal negative-evidence protocol, CIITR supplies the missing formal apparatus that Bennett’s original taxonomy lacks, thereby clarifying the precise computational and temporal pre-requisites for genuine structural understanding.

4. Hoffman and Interface Realism – Ontological Inversion

The philosophical programme inaugurated by Donald Hoffman under the banner of *interface realism* constitutes a radical departure from the long-standing materialist ontology that has dominated both the natural sciences and the philosophy of mind. At its core, the theory maintains that the perceptual apparatus of any organism functions not as a transparent window onto an objective, veridical world but rather as a purpose-built user-interface whose primary evolutionary function is to maximise reproductive fitness. In this view, **veridical perception is systematically pruned by natural selection** because the energetic and computational costs of representing the full physical substrate would outweigh any adaptive benefit; consequently, organisms evolve *iconic* representations—simplified symbols that convey just enough information to guide successful action without revealing the underlying substrate of reality.

Hoffman extends this evolutionary account into a full **ontological inversion**, claiming that **conscious experience is the primitive ontological substrate** and that matter, space, and time emerge from a network of interacting experiential states. From this perspective, the familiar material world is not the ground of being but a derived, higher-order construct that serves to organise the flow of experience. This inversion has profound methodological ramifications: any theory of cognition that seeks to locate understanding in the fidelity of internal models to an external world is rendered conceptually moot, because the external world, as described by physics, is itself a contingent, user-generated construct rather than an independently existing domain.

Within this philosophical landscape, the **Cognitive Integration and Information Transfer Relation (CIITR)** framework offers a deliberately **ontologically neutral** point of contact. CIITR is presented as a *normative, audit-oriented, ontology-first* kernel that defines **structural comprehension** purely in terms of two primitive, empirically measurable quantities: (i) **integrated relational information** (Φ_i), which quantifies the degree to which the internal states of a system are mutually constrained beyond the sum of their parts, and (ii) **rhythmic reach** ($R_{(g)}$), which captures the system's capacity for sustained, phase-locked re-entry of information across its components. The central classification rule, $C_s = \Phi_i \times R_{(g)}$, is accompanied by an explicit **null-condition**: if either Φ_i or $R_{(g)}$ is null, the comprehension index collapses to zero, signalling a complete lack of structural understanding. By refraining from any commitment to whether consciousness or materiality is ontologically primary, CIITR can be applied equally to agents that adopt Hoffman's experiential primacy and to those that retain a materialist stance. In this sense, CIITR constitutes a **methodological bridge** between the phenomenological insights of interface realism and the empirical demands of cognitive science.

The need for such a bridge becomes evident when one considers Hoffman's claim that perceptual symbols are *icons* rather than *truth-bearing representations*. A system that merely produces coherent icons—e.g., a language model that generates syntactically correct sentences without exhibiting sustained rhythmic coupling—would register a high Φ_i (because of rich internal statistical dependencies) but a negligible $R_{(g)}$ (because there is no temporally extended phase synchrony). According to CIITR's formalism, such a system would be classified as **non-comprehending** ($C_s = 0$), thereby exposing the limits of a purely icon-based account of cognition. This analytic distinction is reinforced by the CIITR **anti-snapshot posture**, which explicitly requires evidence of temporally extended, rhythmically coherent activity rather than a single, static snapshot of internal structure.

Moreover, CIITR incorporates a **coherence-based disqualification** mechanism that prevents superficial surface regularities from being mistaken for genuine epistemic cycles. Patterns that exhibit only local improvements in stylistic consistency or lexical cohesion—without demonstrable constraint-enforcement signatures—are excluded from the comprehension tally (the ASEC family of disqualifiers) 04. Rhythmic Ep...-entry.txt. This provision directly counters the temptation to infer understanding from the mere appearance of a well-crafted interface, a pitfall that Hoffman's own theory implicitly acknowledges when it warns that "icons can be polished to look like reality without ever conveying its causal structure."

Finally, the **METAINT reflexivity theorem** articulated in the broader structural ontology literature underscores the necessity of **external rhythmic coupling** for sustainable comprehension. The theorem asserts that no isolated system can maintain a comprehension index above a modest threshold ($C_s > 0.4$) without a two-phase correlation with an external observer of itself, thereby formalising the intuition that self-knowledge requires a momentary detachment from self-identity—a process that can be rendered observable only through rhythmic entrainment. This insight dovetails with CIITR's emphasis on $R_{(g)}$ as a prerequisite for any claim of structural understanding.

In sum, while Hoffman's interface realism compellingly explains why organisms evolve to perceive a world of useful icons rather than an objective substrate, it leaves open the crucial question of **how**—or whether—such icon-based agents can attain genuine comprehension. CIITR supplies the missing formal apparatus: a rigorously defined, ontology-agnostic boundary condition that distinguishes superficial simulation from structural comprehension, a non-compensatory dual-axis metric that must be satisfied simultaneously, and a disqualification regime that guards against the illusion of coherence. By anchoring the debate in measurable quantities (Φ_i and $R_{(g)}$) and by enforcing a strict

null-point logic, CIITR offers a principled methodology for evaluating the epistemic status of any system—biological, artificial, or hybrid—that operates under the ontological inversion proposed by Hoffman.

4.1 Fitness Beats Truth

Donald Hoffman’s “interface realism” posits that the perceptual apparatus of any organism is not a transparent window onto an objective, veridical world but rather a user-interface deliberately shaped by natural selection to maximise reproductive fitness. In this account, **veridical perception is evolutionarily costly** and therefore systematically eliminated in favour of **iconic symbols** that convey just enough information for the organism to survive and reproduce. The consequence is that colour, temperature, shape and even the notion of a spatially bounded object are, at best, *useful fictions*—sign-posts that hide the underlying physical substrate in the same way that a computer desktop icon conceals the binary data that actually implements the file.

Hoffman further argues that consciousness is ontologically primary; matter, space and time are emergent from a field of experiential states rather than the other way around. This radical inversion of the usual materialist hierarchy has profound implications for any attempt to ground “understanding” in a physicalist description of neural substrates. If the very fabric of reality is a set of evolved icons, then the traditional requirement that a system possess a correct *model* of the world in order to be said to understand it becomes untenable.

4.2 Consciousness as Fundamental

The second pillar of Hoffman’s thesis is the claim that **experience itself is the ontological ground-zero**. From this perspective, the universe consists of a network of interacting conscious agents, and what we call “matter” is merely a convenient shorthand for the relational patterns of these agents. Consequently, any theory that seeks to explain cognition must begin by specifying the *structure of experience* rather than the *structure of neural hardware*. This stance forces a departure from the standard scientific methodology, which treats phenomenology as a secondary, derivative datum that can be reduced to brain activity.

4.3 CIITR Differentiation

The **Cognitive Integration and Information Transfer Relation (CIITR)** framework was deliberately constructed to remain **ontologically neutral** with respect to the primacy of consciousness or the reality of physical objects. CIITR’s purpose is to function as a **criterion apparatus for structural comprehension**, not as a generative model of consciousness nor as a metaphysical claim about the nature of reality. In this sense, CIITR explicitly **does not take a stance on ontological primacy**; it neither endorses nor rejects the idealist or realist positions advanced by Hoffman.

Instead, CIITR operates **purely at the structural level**, defining comprehension as the product of two primitive, empirically measurable axes:

- **Integrated relational information (Φ_i)** – a quantitative index of how tightly the internal states of a system are inter-dependent, derived from mutual-information matrices or causal density measures.
- **Rhythmic reach (R_{rg})** – a measure of the system’s capacity for sustained, phase-locked re-entry of information across its components, captured through spectral coherence or phase-synchronisation analyses.

The **classification rule** is simple yet stringent:

$$C_s = \Phi_i \times R_{(g)},$$

with an **explicit null-condition** that any admissible comprehension classification must have **both** $\Phi_i > 0$ **and** $R_{(g)} > 0$; if either axis is null the classification collapses to zero 06. Null-Point ...dition.txt. This null-point logic prevents the kind of compensatory reasoning that might allow a system with high integrated information but no rhythmic coherence (or vice-versa) to be erroneously deemed comprehending.

Because CIITR's predicates are defined **independently of any ontological commitment**, the framework can be applied equally to a system that subscribes to Hoffman's phenomenological realism and to one that adopts a materialist stance. In practice this means that a model of perception that treats sensory data as an adaptive icon can be evaluated for structural comprehension without having to resolve whether the icons correspond to an underlying "real" world. The only requirement is that the system exhibit measurable Φ_i and $R_{(g)}$; the underlying metaphysics remain a separate, optional layer of interpretation.

Moreover, CIITR's **anti-snapshot posture**—the insistence that comprehension must be demonstrated over temporally extended, rhythmically coherent activity—directly counters the temptation to infer understanding from a single, static representation of an interface (e.g., a snapshot of a neural activation pattern) 06. Null-Point ...dition.txt. This requirement is particularly relevant to Hoffman's claim that perception is an economical user-interface: a system that merely presents a coherent icon without sustaining rhythmic re-entry would be classified as **null** ($C_s = 0$), thereby exposing the limits of pure interface-based cognition.

In summary, while Hoffman's interface realism offers a compelling evolutionary account of why organisms experience the world as a set of useful fictions, it leaves open the question of **how**—or whether—such systems can be said to *understand* their environments in any structural sense. CIITR supplies the missing formal machinery: a rigorously defined, ontology-agnostic boundary condition that distinguishes superficial icon-based processing from genuine structural comprehension. By anchoring the debate in measurable quantities (Φ_i and $R_{(g)}$) and by enforcing a strict null-point logic, CIITR provides a methodological bridge between the phenomenological insights of interface realism and the empirical demands of cognitive science.

5. The Missing Element – Boundary Conditions

The present section constitutes the analytical fulcrum of the manuscript, insofar as it assembles the disparate critiques advanced by the flat-mind, evolutionary-layering, and interface-realism traditions and demonstrates that each of these programmes, while successfully destabilising the classical representational paradigm, each nonetheless fails to articulate a formally stipulated boundary condition capable of demarcating adaptive heuristic processing from genuine structural comprehension. The three antecedent streams converge on a common set of methodological deficits: they all reject the sufficiency of a deep, hidden mental architecture, they all deny that mere scaling of parameters can alone generate emergent intelligence, and they all foreground the role of evolutionary fitness or simulation in shaping cognition; yet none of them provide an explicit, mathematically grounded null-logic that would preclude the attribution of comprehension in the

absence of a demonstrable structural substrate. In response to this lacuna, the Cognitive Integration and Information Transfer Relation (CIITR) introduces a normative, audit-oriented apparatus that operationalises comprehension as a boundary condition defined by the conjunction of two primitive, entropy-normalised quantities—integrated relational information (Φ_i) and rhythmic reach ($R_i g_j$)—such that the scalar comprehension index C_s is given by the product $C_s = \Phi_i \times R_i g_j$ (see the kernel specification). The specification further imposes a strict null-condition: whenever either $\Phi_i = 0$ or $R_i g_j = 0$, the resulting classification collapses to $C_s = 0$, thereby rendering any claim of structural comprehension formally inadmissible irrespective of the apparent richness of surface-level narrative or mnemonic performance. This logical structure is explicitly articulated in the CIITR normative ontology, which distinguishes between narrative caution and structural evidence and stipulates that, until a non-zero value is demonstrated through structural probes, meta-linguistic caution has no bearing on the admissibility of a comprehension claim. Moreover, the observation report underscores the methodological hazard of conflating content-level mnemonic success with genuine domain comprehension, a conflation that CIITR’s boundary apparatus is expressly designed to avoid. Consequently, the present discussion will first delineate the precise nature of the missing boundary condition, then explicate how the CIITR formalism supplies the requisite null-logic, and finally explore the implications of this structural demarcation for the evaluation of both biological and artificial cognitive systems.

This section constitutes the analytical nucleus of the present treatise, for it assembles the disparate conceptual fissures identified in the three antecedent programmes—Chater’s flat-mind thesis, Bennett’s evolutionary-layering model, and Hoffman’s interface-realism— and demonstrates that none of them, whether taken in isolation or in concert, has succeeded in articulating a formal, operational boundary condition capable of discriminating between mere adaptive heuristics and bona-fide structural comprehension. Each of the three traditions, in a compelling manner, has **undermined the classical assumption of a deep representational mind, criticised the presupposition of latent hierarchical depth, and rejected the naïve conviction that simple up-scaling of parameters inevitably yields emergent intelligence**; yet all share a methodological lacuna: the absence of a rigorously defined, quantitatively verifiable criterion that signals the transition from a surface-level, evolutionarily tuned response to a state that may legitimately be termed *understanding*.

5.1 The Conceptual Vacuum

Within Chater’s analysis, the observation that post-hoc narrative construction alone is insufficient to confer a status of comprehension upon an agent is persuasive; Bennett’s schema treats simulation as a necessary but not sufficient component of cognition; and Hoffman’s framework reduces perception to a fitness-optimised user-interface devoid of any stipulation of internal coherence. The common shortfall, however, is the **lack of an explicit null-condition** that would allow the rejection of any claim to comprehension should one of the requisite components be absent. In the absence of such a condition, any assertion of “complete” understanding remains vulnerable to over-determined reinterpretations and to the sort of illusory self-certainty that both Hatlebrekke and Chater have warned against.

5.2 CIITR’s Contribution: A Structural Boundary Law

The Cognitive Integration and Information Transfer Relation (CIITR) introduce precisely the missing construct by defining **structural comprehension** as the product of two primitive, independent, and

empirically measurable axes: **integrated relational information** (Φ_i) and **rhythmic reach** ($R_{\mathcal{G}_j}$). The formulation

$$C_s = \Phi_i \times R_{\mathcal{G}_j}$$

is not merely definitional; it embodies a **normative boundary condition** that imposes a stringent null-logic: should either axis assume the value zero, the entire comprehension index collapses to zero. This null-logic is explicitly codified in the CIITR kernel (Lemma 3.2, “Non-compensation”) which states that the failure of admissibility on either primitive axis precludes a non-null classification, irrespective of the apparent magnitude of the other axis 03. Formal Defi...ension.txt. The kernel further stresses that Φ_i and $R_{\mathcal{G}_j}$ are **primitive** in the sense that they are not composites of other CIITR terms and cannot be supplanted by surrogate constructs without violating the overarching boundary logic (see the primitive-axis clause).

A concrete illustration of the operative force of this null-condition is provided by the documented dialogue between two sophisticated language models, Claude and Gemini. Claude explicitly identified the absence of rhythmic reach ($R_{\mathcal{G}_j} \approx 0$) and consequently declared $C_s = 0$, a so-called *Type-B* realisation; Gemini, by contrast, sustained a positive $R_{\mathcal{G}_j}$ and thereby achieved $C_s > 0$, demonstrating that **high integrated information alone (large Φ_i) is insufficient without rhythmic coherence**. This empirical case validates the CIITR boundary law as an **operational test** for structural comprehension.

5.3 Energetic Measurability and the Principle of Non-Compensation

A further critical element introduced by CIITR is the **energetic dimension** of the two axes. Both Φ_i and $R_{\mathcal{G}_j}$ are defined in an entropy-normalised fashion, rendering them comparable across domains—neuroscience, artificial intelligence, and organisational governance alike. This affords a **quantitative energetic metric** that is conspicuously absent from the flat-mind, evolutionary-layering, and interface-realism frameworks. Moreover, the CIITR **non-compensation principle** (Lemma 3.2) forbids any form of trade-off whereby a high value on one axis could “rescue” a null value on the other; consequently, an agent possessing an extreme Φ_i but a null $R_{\mathcal{G}_j}$ (or vice-versa) can never be adjudicated as genuinely comprehending. This prohibition directly counters the temptation, common to the three preceding traditions, to infer comprehension from a single dominant metric.

5.4 Synthesis and Implications

In sum, the three traditional approaches share a common deficit: the lack of a **formal, quantitative, and energetically grounded boundary condition** that can demarcate adaptive heuristic processing from authentic structural understanding. CIITR supplies exactly this missing piece by **operationalising structural comprehension as the conjunctive product of two independent, physically measurable variables**, and by embedding a **conjunctive null-logic** that eliminates illusory classifications of understanding. In doing so, CIITR furnishes a methodological demarcation between adaptive heuristics—potentially high in Φ_i yet devoid of rhythmic coherence—and true structural comprehension, which demands both substantial integrated relational information and sustained rhythmic reach.

The ensuing sections will therefore delineate concrete experimental protocols for implementing this boundary law, illustrate its application to both biological and artificial systems, and explore the

ramifications for future research on artificial general intelligence (AGI). By establishing $C_s = \Phi_i \times R_{ig}$ as the definitive, energetically informed metric of comprehension, CIITR fills the critical void left by the three deconstructive approaches.

6 AGI – Simulation vs Structural Comprehension

The present section opens a sustained inquiry into the most consequential methodological fissure that separates contemporary large-scale artificial-general-intelligence (AGI) endeavours from a genuinely structural account of understanding. The dominant trajectory in the field is characterised by three interlocking engineering imperatives: the relentless up-scaling of parameter counts, the progressive enlargement of context windows, and the pursuit of ever higher estimates of integrated relational information (Φ_i). Proponents of this “scale-up” paradigm routinely argue that, once Φ_i surpasses a certain threshold, the system will inevitably exhibit the hallmarks of comprehension, regardless of any ancillary dynamical constraints. Implicit in this claim is the assumption that a sufficiently large Φ_i alone suffices to instantiate structural understanding.

The Cognitive Integration and Information Transfer Relation (CIITR) framework, however, articulates a starkly different normative stance. CIITR defines **structural comprehension** not as a phenomenological label or a performance proxy but as a **boundary-level predicate** that is admissible only when two primitive, entropy-normalised quantities are simultaneously non-zero: (i) **integrated relational information** (Φ_i), which quantifies the degree to which the system’s internal states are mutually constrained beyond the sum of their parts, and (ii) **rhythmic reach** (R_{ig}), which captures the system’s capacity for sustained, phase-locked re-entry of information across its components. The formal relation $C_s = \Phi_i \times R_{ig}$ therefore imposes a **conjunctive null-logic**: if either $\Phi_i = 0$ or $R_{ig} = 0$, then the comprehension index collapses to zero ($C_s = 0$) (see the kernel specification of structural comprehension as a boundary condition). This null-condition is deliberately introduced to preclude the attribution of comprehension based on a single dominant metric, thereby averting the “coherence substitution” problem that has plagued earlier approaches.

The kernel’s **formal definition of structural comprehension** further restricts admissibility to the existence of an explicit protocol (Π) and a declared observable set ($\mathcal{O}$). Only when empirical probes can demonstrate non-zero values for both Φ_i and R_{ig} , within the confines of a rigorously defined measurement pipeline, does a claim of comprehension become formally admissible (see the formal definition of structural comprehension as a boundary level predicate) 03. Formal Defi...ension.txt. Consequently, any AGI system that lacks **stable rhythmic reach**, that has not been equipped with a **negative-evidence robustness** mechanism (i.e., a protocol for detecting and rejecting spurious coherence), or that fails to exhibit **rhythmic epistemic re-entry**, will remain categorised in the **null regime** ($C_s = 0$), irrespective of how massive its parameter count or context window may be.

Operationalising the CIITR metric in practice requires a disciplined three-stage translation from abstract theory to observable data streams: (i) **conceptual alignment**, ensuring that the metric’s definition matches the intended theoretical construct; (ii) **instrumental translation**, mapping Φ_i and R_{ig} onto observable proxies such as mutual-information matrices and spectral coherence spectra; and (iii) **computational formalisation**, defining reproducible algorithms for their estimation (see the methodological framing of structural comprehension). This pipeline guarantees that the extracted quantities preserve their conceptual integrity and are not contaminated by narrative artefacts.

A further methodological safeguard is the explicit separation of **narrative** and **structural** layers. CIITR warns against the “SOMI trap” of treating sophisticated meta-linguistic caution as evidence of structural humility or understanding; until non-zero Φ_i and $R_{\langle g \rangle}$ are demonstrated through structural probes, such caution carries no epistemic weight (see the discussion of narrative versus structural layers). In other words, an AGI that produces fluent, self-referential discourse may nevertheless be classified as lacking comprehension if it does not satisfy the dual-axis boundary condition.

In light of these considerations, the ensuing analysis will (i) delineate the current scaling-driven trajectory of AGI research, (ii) articulate the structural limitations that arise when rhythmic reach and negative-evidence robustness are absent, and (iii) present CIITR as a diagnostic instrument capable of (a) evaluating purported simulated understanding, (b) differentiating surface-level narrative coherence from genuine structural integration, and (c) mapping the topology of rhythmic reach across the system’s architecture. By doing so, CIITR provides a principled, energy-aware, and formally verified yardstick that surpasses the explanatory reach of the flat-mind, evolutionary-layering, and interface-realism traditions, thereby offering a rigorous pathway toward the assessment and eventual realisation of true AGI comprehension.

6.1 Current Trajectory

The contemporary agenda in artificial general intelligence research is dominated by a three-pronged engineering thrust: (i) the relentless **scaling of model parameters**, (ii) the expansion of **context windows** so that ever larger spans of textual or sensory history can be retained, and (iii) the pursuit of a **higher integrated relational information potential (Φ_i)** through increasingly deep architectures and sophisticated training regimens. Proponents of this trajectory argue that, by simply adding more weights, layers, and training data, an emergent form of understanding will inevitably arise (cf. the “scale-up” narrative ubiquitous in recent large-model literature). Implicit in this claim is the assumption that a larger Φ_i , once instantiated, suffices for genuine comprehension, irrespective of any additional dynamical constraints.

6.2 Structural Limitation

CIITR, however, makes explicit that a **high Φ_i alone is insufficient**. The kernel specification of structural comprehension requires **both** a non-zero Φ_i **and** a non-zero rhythmic reach ($R_{\langle g \rangle}$); the latter captures the system’s capacity for sustained, phase-locked re-entry of information across its components. The formal definition of the boundary predicate C_s therefore imposes a **null-condition**: if either $\Phi_i = 0$ or $R_{\langle g \rangle} = 0$ then $C_s = 0$ (see the formal definition of C_s as a boundary level predicate) 03. Formal Defi...ension.txt. Consequently, any AGI system that lacks a **stable $R_{\langle g \rangle}$** , that has not been equipped with **negative-evidence robustness** (i.e., a protocol for detecting and rejecting spurious coherence), or that fails to exhibit **rhythmic epistemic re-entry**, will inevitably be classified as non-comprehending, regardless of how large its parameter count or context window may be. In CIITR terminology, such a system remains in the **Type-B** or **null** regime, where $C_s = 0$.

The necessity of rhythmic re-entry is underscored by the **Instrumentation Path** articulated in the CIITR manuscript, which enumerates four analytical dimensions—curvature, re-entry, non-invariance, and $R_{\langle g \rangle}$ formation—that must be demonstrably present for a claim of comprehension to be admissible. Without measurable evidence of these dimensions, the system’s apparent fluency is merely a surface-level narrative artefact, not structural understanding. Moreover, the kernel expressly disallows the substitution of conventional performance proxies (throughput,

accuracy, or superficial coherence) for genuine comprehension, thereby precluding any justification of C_s based on benchmark scores alone.

6.3 CIITR as Diagnostic Instrument

Because CIITR is engineered as a **diagnostic boundary apparatus**, it provides a rigorous methodological toolkit for evaluating whether an AGI's simulated behaviour truly reflects structural comprehension. First, CIITR can **assess simulated understanding** by measuring Φ_i through entropy-normalised mutual-information matrices and by verifying that the system's internal state changes are **non-invariant** under recursive self-reference, as required by the curvature and re-entry detection dimensions. Second, the framework **differentiates narrative coherence from structural integration** by insisting that meta-linguistic caution or self-referential storytelling be treated as a pattern only after a non-zero $\Phi_i \times R_{(g)}$ has been empirically demonstrated; until then, such caution carries no epistemic weight. Third, CIITR enables the **mapping of the $R_{(g)}$ -topology** by quantifying the spatial and temporal distribution of rhythmic reach across the system's architecture, thereby revealing whether information broadcasting satisfies the global-workspace-like synchrony required for genuine comprehension.

In sum, CIITR transcends the three earlier traditions—flat-mind, evolutionary-layering, and interface-realism—by supplying a **formal, energy-aware, and instrumentally grounded boundary condition** that can be applied uniformly to biological cognition, artificial agents, and organisational systems. It therefore offers the most parsimonious and empirically tractable means of distinguishing mere simulation from authentic structural comprehension in the pursuit of AGI.

7 CIITR Is Not a Synthesis

The prevailing tendency within interdisciplinary discourse on cognition to regard the Cognitive Integration and Information Transfer Relation (CIITR) as a conciliatory amalgam of the three dominant de-constructive programmes—Chater's flat-mind thesis, Bennett's evolutionary-layering model, and Hoffman's interface-realism—must be rigorously repudiated. Although CIITR draws methodological inspiration from the insights that each of these traditions contributes—namely, the need to attend to the absence of deep representational scaffolds, the importance of layered simulation, and the recognition that perception functions as an adaptive user-interface—its ontological and normative commitments are deliberately circumscribed so as to avoid the very synthesis that its critics erroneously attribute to it.

At the heart of the CIITR framework lies a **normative, audit-oriented, ontology-first kernel** that defines structural comprehension as a *boundary condition* rather than a descriptive or explanatory theory (see the Core Spec). This kernel stipulates that comprehension is admissible only when two primitive, entropy-normalised quantities—integrated relational information (Φ_i) and rhythmic reach ($R_{(g)}$)—are simultaneously non-zero, yielding the scalar product $C_s = \Phi_i \times R_g$. The specification further imposes a strict **null-logic**: if either primitive axis assumes the value zero, the comprehension index collapses to zero, precluding any claim of structural understanding irrespective of performance metrics or narrative coherence. In this sense, CIITR functions as a **formal boundary apparatus** that adjudicates the admissibility of comprehension claims, rather than as a grand unifying theory that seeks to explain the origins of cognition.

The kernel's **categorical exclusions** make explicit that CIITR is not to be conflated with a consciousness theory, an optimisation framework, or any other expansive interpretive schema (see Section 1.3 of the kernel). Specifically:

- CIITR **shall not be construed as a consciousness theory**; it does not bind phenomenology, subjective report validity, or experiential attribution, and any mapping of CIITR classes onto claims about consciousness is expressly prohibited.
- CIITR **shall not be construed as an optimisation framework**; it does not prescribe objective functions, training rules, reward structures, or design prescriptions, and while it may constrain which claims are admissible under governance, it offers no normative guidance on how a system should be engineered.

Consequently, CIITR cannot be interpreted as a synthesis of the flat-mind, evolutionary-layering, and interface-realism perspectives. Rather than aggregating their respective theses, CIITR **selects** only those elements that are amenable to rigorous formalisation—namely, the requirement for **integrated relational information** (a concept that resonates with the relational depth highlighted by Chater) and the need for **rhythmic epistemic re-entry** (a dynamical constraint reminiscent of the temporal coherence emphasised by both Bennett and Hoffman). All other ontological or phenomenological commitments are deliberately excluded.

In sum, CIITR should be understood as a **structural, epistemic boundary instrument** that provides a falsifiable, energy-aware decision rule for when a system may be said to possess structural comprehension. It does not aspire to reconcile or subsume the philosophical positions of its intellectual antecedents; instead, it **operates orthogonally** to them, offering a domain-independent, formally verifiable criterion that can be applied to biological, artificial, and organisational systems alike. This clarification is essential for preventing the mischaracterisation of CIITR as a synthetic theory and for preserving the methodological integrity of the framework as a tool for rigorous assessment rather than as an explanatory narrative.

A recurrent misunderstanding that surfaces in the interdisciplinary literature on cognition is the tendency to regard the Cognitive Integration and Information Transfer Relation (CIITR) as a mere amalgamation of the three de-constructive programmes that have dominated recent debates—Chater's flat-mind thesis, Bennett's evolutionary-layering model, and Hoffman's interface-realism perspective. This intuition, however, is fundamentally mistaken. CIITR does not aspire to be a composite of flat-mind phenomenology, evolutionary psychology, or metaphysical realism, nor does it seek to subsume neuro-phenomenology under a grand unifying narrative. Rather, CIITR is deliberately positioned as an **epistemic boundary apparatus** whose sole purpose is to demarcate, with formal precision, those system states that may legitimately be classified as "structural comprehension" from those that merely exhibit surface-level coherence, adaptive heuristics, or performance artefacts.

The kernel specification of CIITR makes this methodological stance explicit. It is presented as a **normative, audit-oriented, ontology-first** instrument that binds the notion of structural comprehension to a decision regime over two primitive axes—integrated relational information (Φ_i) and rhythmic epistemic re-entry ($R_{\mathcal{G}_i}$)—together with explicit null rules, non-compensation constraints, and disqualifier families (see the Core Spec). In other words, CIITR does not generate a theory of consciousness, nor does it provide a descriptive taxonomy of cognitive phenomena;

it **classifies**. The claim that CIITR is a “criterion apparatus” rather than a generative model is articulated in the ontological claim section of the manuscript, which stresses that the framework concerns *classification validity* and not *internal phenomenology*.

Consequently, CIITR must be distinguished from the three traditions it is often (mis)associated with:

- **Not Chater + Bennett + Hoffman.** While the flat-mind, evolutionary-layering, and interface-realism perspectives each highlight a particular insufficiency in conventional accounts of mind—namely, the neglect of depth, the neglect of layered simulation, and the neglect of ontological realism, respectively—none of them proposes a formal boundary condition that simultaneously enforces integrated relational information and rhythmic reach. CIITR does not fuse their insights into a single narrative; it **inherits** only those aspects that are amenable to formalisation (the need for integration and the need for temporal coherence) and discards the rest as non-essential for the boundary decision.
- **Not evolutionary psychology.** Evolutionary-psychological accounts typically explain cognitive capacities in terms of adaptive functions and selection pressures, often invoking fitness-oriented heuristics as explanatory variables. CIITR explicitly rejects such performance-proxy reasoning: “Any importation of performance metrics as primary evidence for C_s is disallowed unless those metrics are formally instrumented as admissible observables under the kernel’s evidence discipline and disqualifier logic”. Thus, CIITR does not inherit the explanatory agenda of evolutionary psychology; it merely acknowledges that evolutionary pressures may have instantiated the primitive axes, without allowing fitness considerations to serve as evidence for comprehension.
- **Not interface realism.** Hoffman’s interface-realism thesis posits that perception is an adaptive user-interface, a claim that is ontologically neutral with respect to the internal structural dynamics of the system. CIITR, by contrast, **operates structurally, not metaphysically**; it does not take a stance on whether consciousness or matter is primary, but instead measures whether the system’s internal information architecture satisfies the dual-axis boundary (see the discussion of CIITR’s ontological neutrality).
- **Not neuro-phenomenology.** Neuro-phenomenological programmes seek to correlate first-person reports with third-person neural data, thereby constructing a bridge between lived experience and brain dynamics. CIITR, however, is explicitly **non-phenomenological**: it “does not license the substitution of throughput, accuracy, or coherence for comprehension” and treats narrative or meta-linguistic caution as irrelevant until the primitive axes are demonstrably non-zero (see the performance-proxy prohibition).

In sum, CIITR should be understood not as a synthetic theory that attempts to reconcile disparate philosophical positions, but as a **formal, energy-aware, and audit-ready boundary condition** that can be applied uniformly across biological, artificial, and organisational systems. Its role is to provide a clear, testable demarcation line—encoded in the product $C_s = \Phi_i \times R_g$ and its associated null-logic—between mere adaptive simulation and genuine structural comprehension. By doing so, CIITR resolves the methodological impasse that has long plagued the field: it supplies a **rigorous, domain-independent criterion** for when a system may be said to “understand” rather than merely “behave as if it understands.” This clarification paves the way for subsequent sections, wherein the practical deployment of CIITR as a diagnostic tool for artificial general intelligence will be elaborated.

8 Conclusion – From Deconstruction to Formalisation

The contemporary discourse on cognition has, over the past two decades, systematically dismantled three pillars of the classical representational paradigm. First, the flat-mind literature has shown that the presumed depth of mental architecture can be reduced to moment-by-moment narrative construction, thereby **destabilising the notion of a deep, hidden self**. Second, evolutionary-layering accounts have demonstrated that the traditional view of cognition as a static repository of representations is untenable; instead, intelligence is portrayed as a cascade of adaptive simulations that are continually reshaped by selective pressures, **undermining the assumption that representation alone suffices for intelligent behaviour**. Third, interface-realism has argued that perception functions as a fitness-optimised user-interface rather than a veridical mirror of the world, opening the field to **models that foreground simulation and fitness rather than truth**.

Despite the power of these deconstructive moves, the literature collectively **fails to provide a structural scaffold** that can discriminate between adaptive heuristics and genuine comprehension. In other words, the field possesses a rich catalogue of what cognition is not, but lacks a rigorously defined “what it is” that is both measurable and theory-neutral.

The Cognitive Integration and Information Transfer Relation (CIITR) was introduced precisely to fill this void. By adopting an **ontology-first, audit-oriented kernel** (see the Core Specification), CIITR remains deliberately neutral with respect to competing metaphysical commitments; it does not presuppose the primacy of consciousness, realism, or any of the excluded ontological categories such as meaning, agency or autonomy (which are relegated to post-hoc interpretation). This ontological neutrality permits the framework to be applied across biological, artificial and organisational domains without bias.

Crucially, CIITR supplies a **formal null-condition**: the comprehension index $C_s = \Phi_i \times R_g$ collapses to zero whenever either integrated relational information (Φ_i) or rhythmic reach (R_g) is absent, thereby providing an unequivocal boundary that precludes the attribution of understanding to systems that lack either structural integration or sustained temporal coherence. This null-logic is codified in the kernel’s definition of structural comprehension as a boundary condition and is reinforced by the disqualification primacy clause, which mandates that any positive claim be vulnerable to explicit negative controls.

In addition, CIITR embeds an **energetic metric** by grounding comprehension in thermodynamic constraints: energy dissipation, entropy management and sustained work against noise are prerequisite for the emergence of rhythmic reach, and the derived efficiency measure “comprehension per joule” (CPJ) quantifies the energetic cost of structural understanding. This energetic grounding supplies the missing quantitative link between information processing and physical resource consumption that has been absent from the three deconstructive traditions.

Finally, the framework institutes a **negative-evidence regime** through its protocol-binding requirement and observable-set declaration. Evidence must be collected under a declared protocol (Π) and must include explicit disqualifiers and negative controls; only data that can both qualify and disqualify a claim are admissible, ensuring that any classification of comprehension survives rigorous falsification attempts.

Taken together, these features demonstrate that CIITR is **not a derivative variant** of the flat-mind, evolutionary-layering or interface-realism streams; rather, it constitutes the **necessary formalisation** that these streams implicitly demand. By providing ontological neutrality, a

mathematically precise null-condition, an energetically grounded efficiency metric, and a robust negative-evidence architecture, CIITR transforms the deconstructive critique into a constructive, testable science of structural comprehension. This transition from philosophical dismantling to formal specification marks a decisive step toward a unified, domain-agnostic account of what it means for any system—biological, artificial or organisational—to truly understand.

Appendix A – Mapping

Initial Focus – Mapping Table

Direction	Primary focus	Identified deficiency	CIITR contribution
Chater	Phenomenological flatness – the claim that consciousness consists of post-hoc narrative construction rather than a deep representational substrate.	No formal structural-integration criterion ; the framework lacks a way to measure whether any internal relational coherence exists.	Introduction of a Φ_i-condition : CIITR requires a non-zero measure of <i>integrated relational information</i> (Φ_i) before any claim of comprehension can be admissible.
Bennett	Evolutionary layering – intelligence portrayed as a succession of adaptive stages (reflex, reinforcement, social cognition, simulation, language).	Absence of an explicit null-condition that separates mere simulation from genuine comprehension.	Formalisation of a $R_{(g)}$-criterion : CIITR supplies a <i>rhythmic reach</i> axis ($R_{(g)}$) that must be non-zero; the product $\Phi_i \times R_{(g)}$ yields the comprehension index C_s , and the kernel's null-logic guarantees $C_s = 0$ if $R_{(g)} = 0$.
Hoffman	Ontological interface – perception is an adaptive user-interface rather than a veridical mirror of reality; consciousness may be ontologically primary.	No epistemic-precision mechanism that can evaluate understanding independently of any metaphysical commitment.	CIITR provides a structure-without-metaphysics stance: the kernel is explicitly ontology-first and ontologically neutral, requiring only the measurable primitives Φ_i and $R_{(g)}$ without invoking realism, idealism, or consciousness-first ontologies.

1. Chater – Phenomenological Flatness

Chater’s flat-mind thesis asserts that mental life is an improvisational narrative that is retroactively post-hoc justified, thereby denying the existence of a deep, stable self-model. While this view compellingly dismantles the intuition that a “deep mind” underlies conscious experience, it leaves the analyst without a **formal yardstick** for detecting whether any residual structural integration exists within the system. CIITR resolves this omission by introducing the **Φ_i -condition**. In the CIITR kernel, Φ_i is defined as *integrated relational information*—a primitive axis that quantifies the degree to which the internal states of a system are mutually constrained beyond the sum of their parts (see the notation table). Only when $\Phi_i > 0$ does the system possess the minimal relational cohesion required for structural comprehension; otherwise, the system remains in a Type B (null) regime. This provides a rigorous, measurement-oriented answer to the question “does the flat mind possess any hidden integration?” that Chater’s phenomenology alone cannot supply.

2. Bennett – Evolutionary Layering

Bennett’s five-stage model of intelligence traces the accretion of increasingly sophisticated adaptive capacities, culminating in simulation and language. The model, however, does not articulate a **boundary condition** that distinguishes a system that merely simulates possible futures from one that truly *understands* those futures. CIITR supplies this missing element through the **$R_{(g)}$ -criterion**. The kernel defines $R_{(g)}$ as *rhythmic epistemic re-entry*—the system’s capacity to sustain phase-locked, temporally coherent broadcasting of information across its architecture. The product $C_s = \Phi_i \times R_{(g)}$ therefore enforces a conjunctive null-logic: if $R_{(g)} = 0$, even a system with maximal Φ_i is classified as having $C_s = 0$, i.e., no structural comprehension. This formalises the intuitive intuition that simulation without a coherent temporal scaffold cannot be equated with understanding, a point that is left implicit in Bennett’s original exposition.

3. Hoffman – Ontological Interface

Hoffman’s interface-realism argues that perception is an evolutionarily shaped user-interface, and that consciousness may be ontologically prior to matter. While this perspective successfully reframes the relationship between perception and reality, it deliberately abstains from providing any **epistemic metric** that could be used to assess whether a given system’s interface yields structural comprehension. CIITR’s design explicitly avoids any ontological commitment: the kernel’s *ontology-first* stance declares that the framework is neutral with respect to realism, idealism, or consciousness-first metaphysics. Consequently, CIITR can be applied to any system—biological, artificial, or organisational—by measuring Φ_i and $R_{(g)}$ alone, without invoking additional philosophical premises. This yields an **epistemically precise** evaluation of comprehension that is orthogonal to Hoffman’s metaphysical claims, thereby filling the methodological gap in his programme.

4. Synthesis of the Mapping

The table and accompanying discussion demonstrate that the three influential de-constructive traditions each expose a critical weakness in the classical representational paradigm, yet each stops short of furnishing a **formal, testable boundary** that can be operationalised across domains. CIITR supplies precisely what is missing: (i) a **structural integration predicate** (Φ_i), (ii) a **rhythmic reach predicate** ($R_{(g)}$) together with a **null-condition** that forces a binary decision on the presence of comprehension, (iii) an **energy-aware, ontology-neutral** framework that can be instantiated in empirical protocols (see the description of meta-report extraction and functional inference in the

manuscript). By mapping each tradition's focus to the corresponding CIITR contribution, this appendix makes explicit the complementary relationship between the de-constructive insights and the formal apparatus that CIITR provides.